

MAT 426, Class 24: Gram-Schmidt and Cauchy-Schwarz Inequality

Definition. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V .

- (1) The **length** (or **norm**) of $\mathbf{v}$ is $\|\mathbf{v}\| =$ _____.
- (2) The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) =$ _____.
- (3) $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if _____.

Example 1. Consider the inner product on $\mathcal{C}[0, 1]$, $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$, from Example 3. Let $f(x) = x$ and $g(x) = 3x - 2$. Compute the following:

- (a) $\|f\|$ (b) $d(f, g)$ (c) $\langle f, g \rangle$

Example 2. Consider the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2$ from Example 1. Sketch the “unit circle” corresponding to this inner product.

Definition.

- An **orthogonal set** of vectors in an inner product space V is a set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of vectors from V such that _____ whenever _____.
- An **orthonormal set** of vectors is an _____ set of _____ vectors.
- An **orthogonal basis** for a subspace W of V is a _____ that is an _____ set. An **orthonormal basis** for a subspace W of V is a _____ that is an _____ set.

Example 3. (*Legendre Polynomials*) Construct an orthogonal basis for $\mathcal{P}_3$ (the set of all polynomials of degree at most 3) with respect to the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$$

by applying the Gram-Schmidt process to the basis $\{1, x, x^2, x^3\}$.

Example 4. (*Chebyshev Polynomials*) Let $\mathcal{P}$ be the vector space of all polynomials.

- (a) Show that $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)w(x) dx$ is an inner product where $w(x)$ is a continuous function.
- (b) Let $w(x) = \frac{1}{\sqrt{1-x^2}}$. Consider the standard basis $\{1, x, x^2, x^3, \dots\}$. Use Gram-Schmidt to construct an orthogonal basis for $\mathcal{P}$

Definition. Suppose that W is a subspace of a vector space V that has an orthogonal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. The **orthogonal projection** $\text{proj}_W(\mathbf{v})$ of a vector $\mathbf{v}$ onto the subspace W is

$$\text{proj}_W(\mathbf{v}) =$$

The **component of $\mathbf{v}$ orthogonal to W** is the vector

$$\text{perp}_W(\mathbf{v}) =$$

CAUCHY-SCHWARZ INEQUALITY Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V . Then,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

with equality holding if and only if $\mathbf{u}$ and $\mathbf{v}$ are scalar multiples of each other.

Proof.